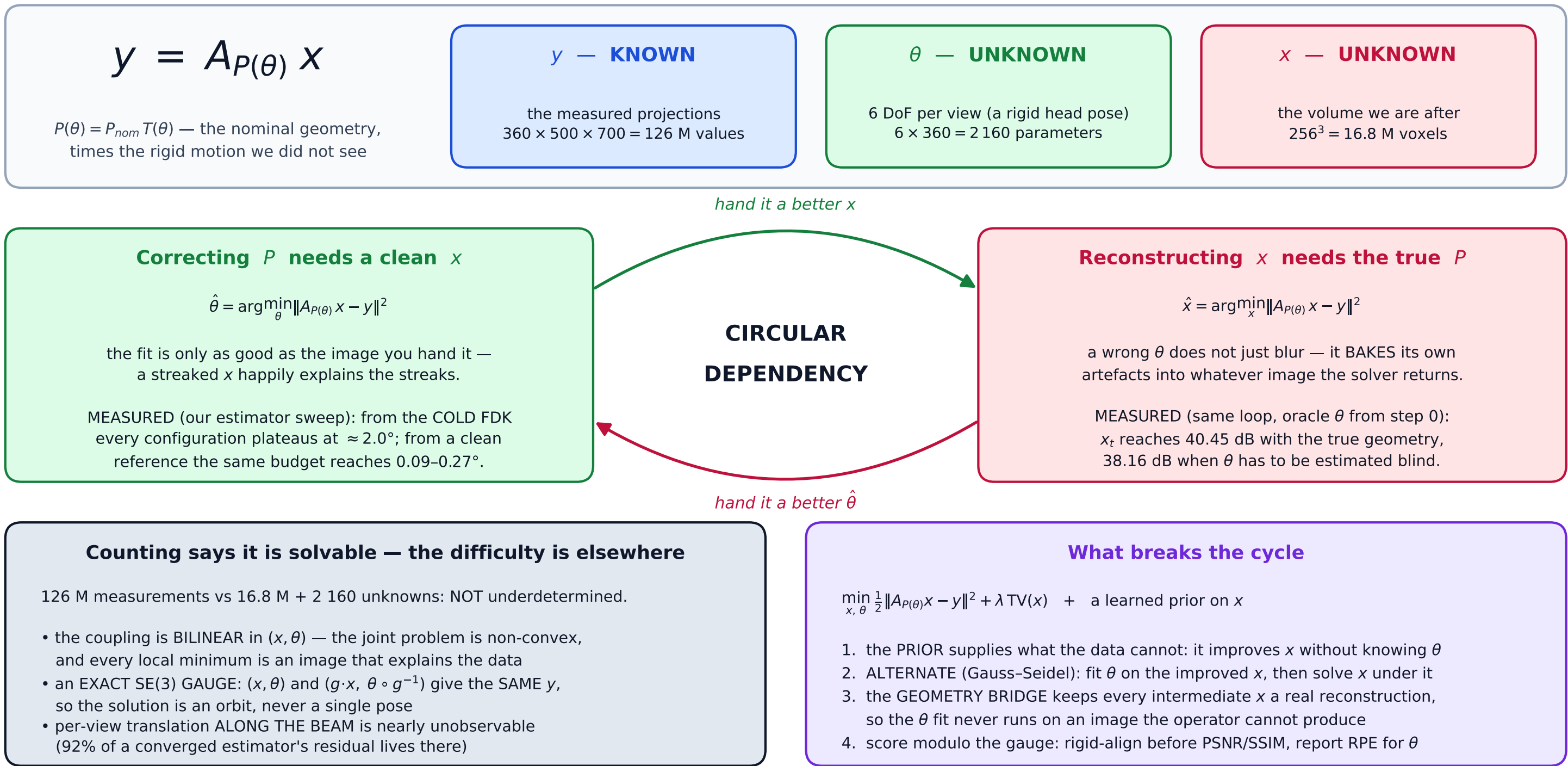


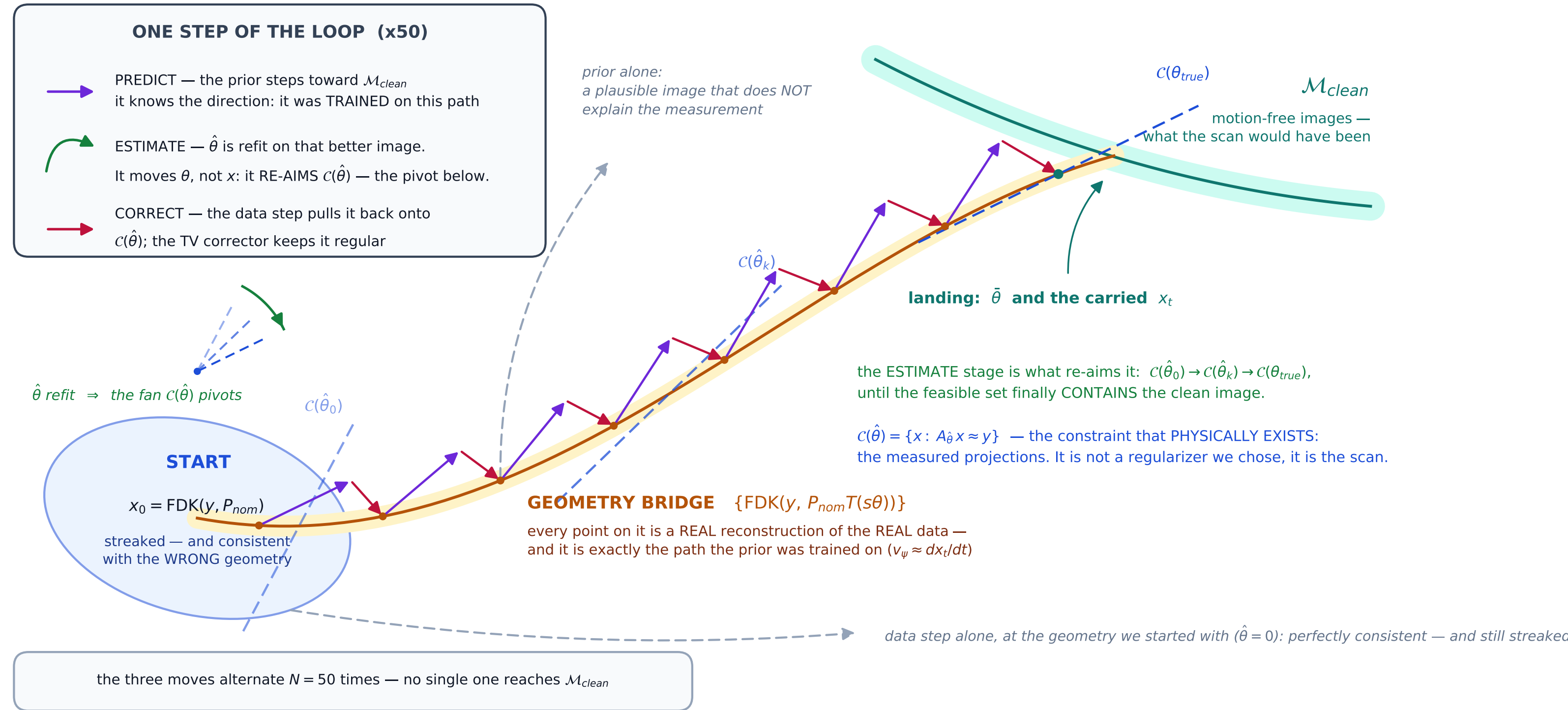
# The problem is BLIND: one measurement, two unknowns

$P$  cannot be corrected without a clean  $x$  — and  $x$  cannot be reconstructed without the right  $P$



## Why the loop works: a learned path to the clean manifold, held on the physics at every step

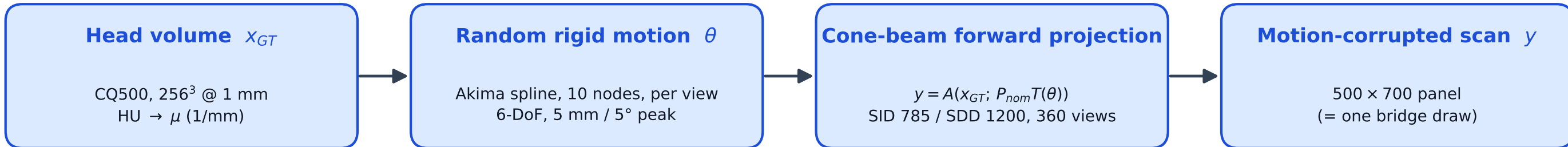
the prior supplies the DIRECTION (it was trained on this very path); the measured data supplies the CONSTRAINT that keeps the direction honest



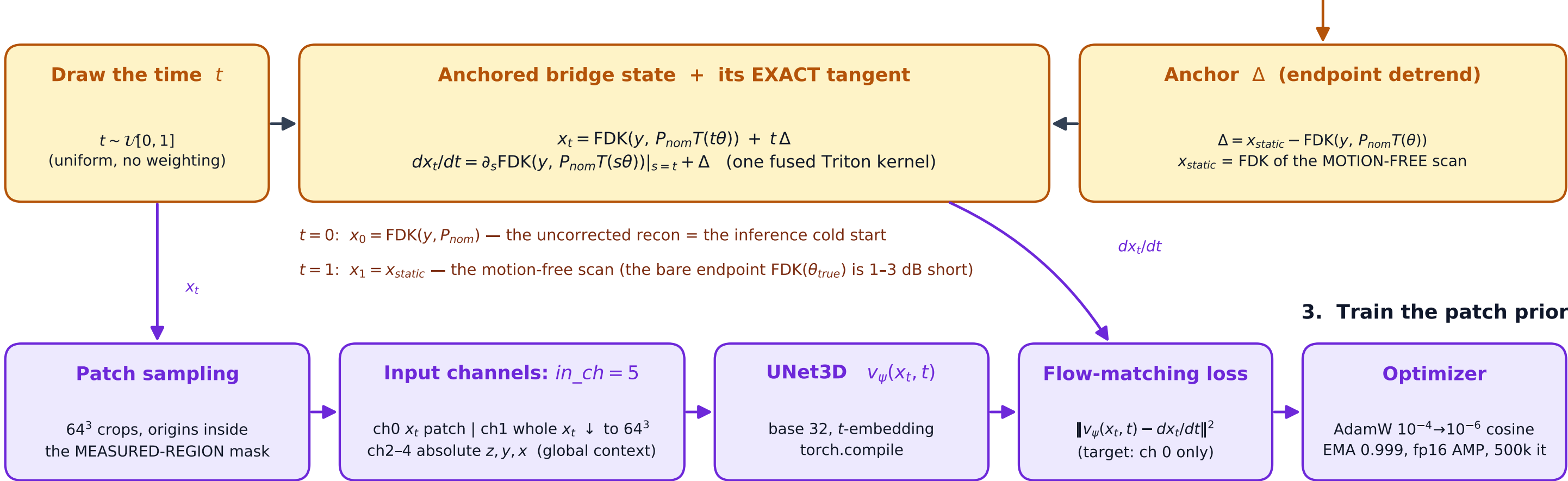
## Training: a flow-matching prior on the GEOMETRY BRIDGE

the prior is trained on exactly the image manifold the inference ODE walks: reconstructions under a PARTIALLY CORRECTED geometry

### 1. Simulate a motion-corrupted scan (CQ500, train split)



### 2. Geometry bridge (whole volume, no\_grad — one fused FDK+tangent pass)



MEMORY SPLIT — the OPERATOR (FDK + its tangent) runs on the FULL volume under no\_grad; the NET only ever sees 64<sup>3</sup> patches.  
A rolling cache of 8 whole-volume bridge draws is refreshed every 12 steps; each batch of 8 patches mixes several draws, so  $t$  varies within a batch.  
Validation = prior-only ODE from the cold start on held-out patients: 25.42 dB / SSIM 0.78 (rigid-aligned).

## Inference: blind rigid-motion correction — a PnP-TV predictor-corrector loop

$\theta$  is UNKNOWN: the image and the motion bootstrap each other, one Euler step of the FM ODE at a time ( $N = 50$  steps)

